Reducing Customer Churn Through Predictive Risk Scoring

A business report on the Telco customer retention model

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0.844	78.3%	53.4%	0.507
ROC-AUC on held-out customers	of churners caught in advance	precision of high-risk flags	ROC-AUC of a random-guess baseline

1. Executive summary

The problem. Customer churn — subscribers cancelling their service — is the largest controllable drag on recurring revenue for a subscription telecom business. Retention teams typically contact customers reactively (after cancellation is already underway) or indiscriminately (spending limited budget on customers who were never leaving). Neither uses the budget efficiently.

The data. 7,043 historical customer records covering demographics, account details (contract, tenure, charges, payment method) and subscribed services.

The model. Four classifiers were compared — a random-guess baseline, Logistic Regression, a Decision Tree and a Random Forest — and the Random Forest, tuned with a systematic hyperparameter search (GridSearchCV), was selected as the final model.

The result. The final model reaches **84.4% ROC-AUC** (95% confidence interval 82.2%–86.5%) on customers it never saw during training, far above the 50.7% a random-guess baseline achieves, which confirms it has learned genuine patterns rather than noise. It catches **78.3% of true churners** in advance (95% CI 73.9%–82.2%), clearing the deployment bar set out in Section 2 with room to spare.

What next. Deploy the model into the monthly retention workflow (Sections 6 and 7), starting with the highest-leverage actions in Section 6 — contract-length incentives and first-six-month onboarding — validated with a small controlled test before scaling company-wide.

2. Business problem

The retention team's core constraint is a limited monthly outreach budget, not a lack of awareness that churn happens. A risk-scoring model lets that budget be spent on the customers where an intervention is most likely to change the outcome, rather than reactively or indiscriminately.

Cost asymmetry behind the model's design

Missing an actual churner (a "false negative") loses that customer's full future revenue with no chance to intervene; flagging a customer who was never leaving (a "false positive") only costs an unnecessary discount or call. Because a missed churner is far more expensive, the model is deliberately tuned to prioritise catching true churners, accepting a modest increase in false alarms as the price of doing so.

Deployment threshold: what makes the model worth shipping

Before building anything, an explicit bar was set: the model is worth deploying only if it reaches a ROC-AUC of at least 0.75 (meaningfully better than the 0.50 a coin flip achieves) and a recall of at least 0.65 (catching at least two-thirds of true churners), both measured on data the model never trained on and using the conservative lower end of a 95% confidence interval rather than a single lucky point estimate. Section 5 confirms that the final model clears both bars.

3. Data and methodology

The model was trained on 7,043 anonymised customer records covering demographics, account details and subscribed services; 26.5% of customers in the historical data had churned. Two additional signals were engineered from the raw data: a count of optional add-on services per customer, and a flag for customers in their first six months — the period Section 4 shows carries the sharpest churn risk. All numeric features were scaled and all categorical features encoded inside a single reproducible pipeline, fit only on a training subset (80% of customers) and evaluated on a held-out 20% test subset it never saw.

Four models were compared: a random-guess baseline (the floor any real model must beat), Logistic Regression, a single Decision Tree and a Random Forest ensemble. The Random Forest was then tuned with a systematic search over its key parameters (tree depth, leaf size, ensemble size) and selected as the final model.

4. Key findings

4.1 Contract length is the single strongest predictor

Month-to-month customers churn at **42.7%**, versus just 11.3% for one-year contracts and **2.8%** for two-year contracts — over nine times more likely to leave than a two-year customer. This holds across every internet service type, which confirms it is a genuine driver rather than a side effect of something else.

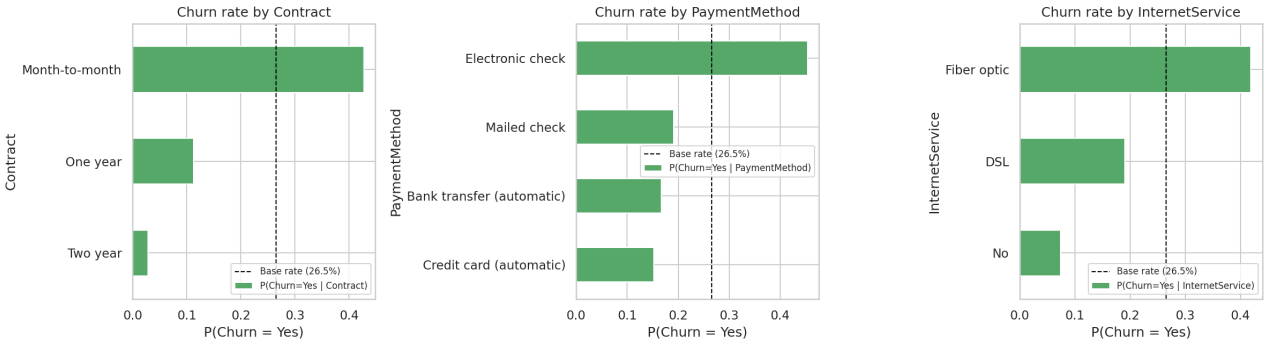


Figure 1 — Churn rate by contract type, payment method and internet service, against the 26.5% company-wide average (dashed line).

4.2 Risk is concentrated in a customer's first six months

Customers in their first six months churn at over twice the rate of established customers. This is also the window where intervention is most likely to change the outcome, which is why it is engineered directly into the model as a feature rather than left implicit.

4.3 Payment method is a strong, actionable secondary signal

Customers paying by electronic check churn at **45.3%**, roughly double the rate of customers on automatic bank transfer or credit card payments. Automatic payment appears to be "sticky": cancelling requires an active decision instead of a payment simply lapsing.

5. Model performance

The final model was evaluated on 1,409 customers held out from training. The figures below include 95% confidence intervals from bootstrap resampling instead of single point estimates, so the business can judge how much the numbers might vary on a different sample of customers.

Metric	Score (95% CI)	What it means for the business
ROC-AUC	0.844 (0.822–0.865)	Probability that the model ranks a true churner above a non-churner — reliably far better than chance (0.507, see baseline below).
Recall (churners caught)	78.3% (73.9–82.2%)	Of all customers who actually churned, the model would flag roughly 78 out of every 100 in advance.
Precision (flag accuracy)	53.4%	Of customers flagged as high-risk, just over half genuinely churn; the remainder still receive a low-cost, low-risk touchpoint.
Accuracy	76.1%	Included for completeness. It is not the primary metric; see the note below.

5.1 Proof the model learned real signal: comparison against a random baseline

A baseline model that ignores every customer feature and guesses churn using only the overall 26.5% base rate scores 50.7% ROC-AUC, statistically indistinguishable from a coin flip. The final model's 84.4% ROC-AUC is a very large, consistent margin above this floor.

Why not simply optimise accuracy? Only 26.5% of customers churn, so a model predicting "will not churn" for everyone would score 76.1% accuracy while catching zero at-risk customers. Recall and ROC-AUC are the metrics that reflect whether the model helps the retention team find customers worth calling.

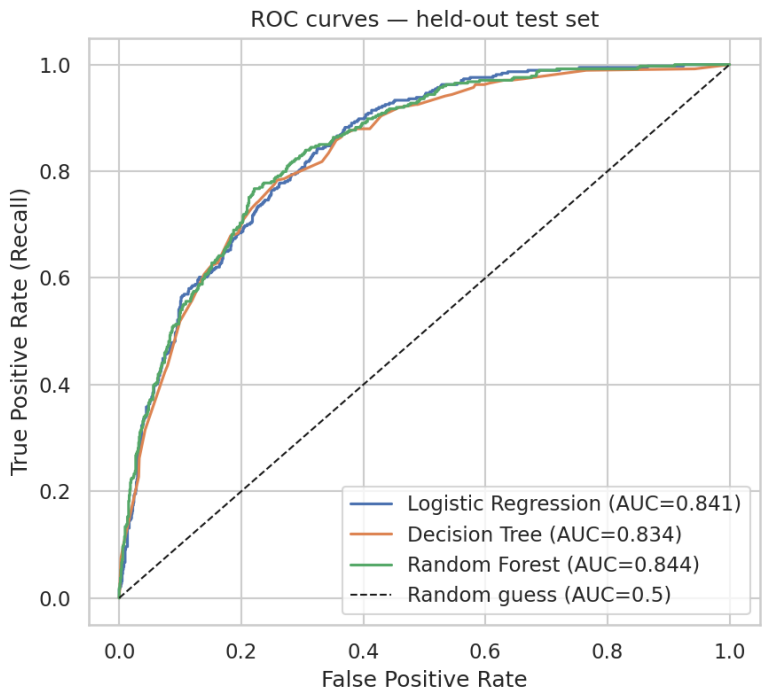


Figure 2 — ROC curves on the held-out test set: the further above the diagonal (random-guess) line, the better the model separates churners from non-churners at every possible risk threshold.

5.2 A fairness check: does the model perform evenly across customer groups?

Recall was checked separately by gender and by senior-citizen status. The gap across gender is small (5.6 percentage points), as expected, since gender has no plausible causal link to churn. The gap across

senior-citizen status is larger (12.8 percentage points): senior citizens churn at a higher underlying rate and the model catches a higher share of them. This is worth surfacing to the business explicitly instead of assuming the score treats every customer group identically; any team acting mechanically on this score should be aware that age groups are not caught at exactly the same rate.

6. Recommendations

1. Prioritise the first six months of the customer lifecycle.
Introduce a structured onboarding and check-in touchpoint at 30, 90 and 150 days for new customers, since this window carries both the highest churn risk and the greatest opportunity to influence the outcome.

2. Incentivise contract upgrades.
Offer month-to-month customers a modest discount or perk for moving to a one- or two-year term. Given the nine-fold churn-rate gap between month-to-month and two-year customers, even a partial shift toward longer terms would materially reduce churn.

3. Encourage automatic payment methods.
Offer a small incentive (for example a one-time credit) for switching from electronic check to automatic bank transfer or credit card. This is a low-cost lever with a large observed association with retention.

4. Deploy the risk score into the monthly retention workflow.
Use the accompanying model (`model.joblib` / `predict.py`) to score the active customer base monthly and route the top-ranked at-risk customers to the retention team ahead of the lower-risk majority. A starting decision threshold of 0.35 (instead of the default 0.50) is recommended to bias toward catching more true churners, consistent with the cost asymmetry in Section 2; tune it further once real per-contact costs are available. Given the fairness finding in Section 5.2, monitor outreach rates by age group to confirm they stay proportionate to actual risk and not just to the raw score.

7. Limitations and next steps

This model is trained on a single historical snapshot and does not yet incorporate ongoing behavioural signals such as recent support tickets or usage trends. The findings are correlational, not causal: it cannot yet be claimed that switching a customer to automatic payment causes them to stay, only that the two are strongly associated.

Before committing significant budget to the recommendations above, run a small controlled test — for example, offer the contract-upgrade incentive to a random sample of eligible customers and compare their churn rate with a control group — to confirm the causal effect before scaling it company-wide. The subgroup gap in Section 5.2 should also be monitored over time as the model is retrained on newer data.

Full technical detail — data cleaning, statistical analysis, model comparison, hyperparameter tuning and code — is available in the accompanying Jupyter notebook and README.